SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Kanish Kumar V	**Reg. No.:**	192524361
Date:		**Duration:**	60 minutes

Code of Conduct

I am Kanish Kumar V (192524361) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kanish Kumar V_____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

Smart Library Seat Reservation and Study Space Management System

AGILE SPRINT PLANNING AND USER STORY WORKSHOP

1. Project Vision and Stakeholders

1.1 Project Vision

The vision of the Smart Library Seat Reservation and Study Space Management System is to develop a user-friendly software application that enables students to find and reserve library study spaces efficiently. The system will manage seat availability, reservations, check-in and check-out activities, and unused reservations. It will help librarians monitor occupancy and improve the utilization of available study spaces.

1.2 Project Objectives

The main objectives of the project are:

1. To display available study spaces and their current status.
2. To enable students to reserve seats for specific time slots.
3. To provide check-in and check-out facilities.
4. To monitor seat occupancy and reservation status.
5. To identify unused reservations and release seats according to library policy.
6. To manage reservation time limits and grace periods.
7. To send reservation confirmations and reminders.
8. To generate seat utilization reports for library administrators.
9. To reduce the time students spend searching for seats.
10. To improve library resource utilization and administrative efficiency.

1.3 Key Stakeholders

Stakeholder	Role and Responsibilities
Students	View available seats, reserve seats, check in, check out, and cancel reservations.
Librarians	Monitor seat usage, assist students, and manage seating-related issues.
Library Administrators	Configure seats, set reservation policies, and review utilization reports.
System Administrator	Maintain the application, manage access, and handle technical issues.
College Management	Review utilization trends and plan library capacity improvements.
Development Team	Analyze requirements, design, develop, test, and maintain the system.
Notification Service Provider	Deliver reservation confirmations, reminders, and alerts.

2. Create Epics and User Stories

An **epic** is a large functionality that can be divided into smaller user stories. A user story describes a feature from the perspective of the person who will use it.

The following epics and user stories are identified for the project.

Epic 1: User Authentication and Access Management

US-01: Student Login

As a student, I want to log in to the system, so that I can access the seat reservation services securely.

US-02: Role-Based Access

As a library administrator, I want users to have role-based permissions, so that only authorized users can access administrative functions.

Epic 2: Seat Availability Management

US-03: View Available Seats

As a student, I want to view available seats by study area, so that I can quickly find a suitable place to study.

US-04: View Seat Details

As a student, I want to view seat details and available time slots, so that I can select a suitable seat and reservation period.

US-05: Manage Seat Information

As a librarian, I want to add, update, or mark seats as unavailable, so that the displayed seat information remains accurate.

Epic 3: Seat Reservation Management

US-06: Reserve a Seat

As a student, I want to reserve an available seat for a specific time slot, so that I can secure a place to study.

US-07: Prevent Double Booking

As a student, I want the system to prevent overlapping reservations for the same seat, so that my reservation remains valid.

US-08: Cancel a Reservation

As a student, I want to cancel my reservation, so that other students can use the seat if I no longer need it.

Epic 4: Check-in, Check-out and Occupancy Monitoring

US-09: Check In

As a student, I want to check in when I arrive at the library, so that the system records my arrival and updates my reservation status.

US-10: Check Out

As a student, I want to check out after completing my study session, so that the seat becomes available to other students.

US-11: Monitor Occupancy

As a librarian, I want to monitor seat occupancy and reservation status, so that I can manage the study areas efficiently.

Epic 5: No-Show Detection and Time Management

US-12: Detect Unused Reservations

As a librarian, I want the system to identify reservations that have not been checked in within the allowed grace period, so that unused seats can be released.

US-13: Enforce Reservation Time Limits

As a library administrator, I want to configure reservation durations and check-in grace periods, so that seat usage follows library policies.

Epic 6: Notifications and Reports

US-14: Reservation Notifications

As a student, I want to receive reservation confirmations and reminders, so that I do not forget my booking or reservation time.

US-15: Seat Utilization Reports

As a library administrator, I want to generate seat utilization reports, so that I can identify peak hours and improve space planning.

3. Define Acceptance Criteria

Acceptance criteria specify the conditions that must be satisfied for a user story to be considered complete. The Given–When–Then format is used to make these conditions clear and testable.

US-01: Student Login

- **Given** a registered student has valid credentials, **When** the student submits the login form, **Then** the system authenticates the student and displays the home page.
- **Given** a student enters invalid credentials, **When** the login form is submitted, **Then** the system displays an error message and does not grant access.

US-03: View Available Seats

- **Given** the system contains seat records, **When** a student opens the availability page, **Then** the system displays seats with their current status.
- **Given** a seat is reserved or unavailable for the selected time slot, **When** the student views that time slot, **Then** the seat is not offered as available for booking.

US-04: View Seat Details

- **Given** a student selects a study area, **When** the student opens a seat's details, **Then** the system displays the seat identifier, area, and available time slots.

US-05: Manage Seat Information

- **Given** a librarian has the required permissions, **When** the librarian adds or updates a seat, **Then** the system saves the changes and reflects them in the availability view.
- **Given** a seat is under maintenance, **When** the librarian marks it unavailable, **Then** the system prevents new reservations for the affected period.

US-06: Reserve a Seat

- **Given** a student is logged in and a seat is available, **When** the student selects a valid time slot and confirms the booking, **Then** the system creates a reservation and displays a confirmation.
- **Given** the selected seat is no longer available, **When** the student attempts to reserve it, **Then** the system rejects the booking and requests another selection.

US-07: Prevent Double Booking

- **Given** a seat already has a confirmed reservation for a time slot, **When** another student attempts an overlapping booking, **Then** the system rejects the conflicting reservation.
- **Given** two students attempt to reserve the same seat simultaneously, **When** both requests are processed, **Then** no more than one conflicting reservation is confirmed.

US-08: Cancel a Reservation

- **Given** a student has an eligible active reservation, **When** the student cancels it, **Then** the system updates the reservation status and makes the seat available for the applicable time slot.

US-09: Check In

- **Given** a student has a valid reservation within the permitted check-in period, **When** the student checks in, **Then** the system records the check-in time and updates the reservation status.
- **Given** a student has no valid reservation, **When** the student attempts to check in, **Then** the system rejects the request and displays an appropriate message.

US-10: Check Out

- **Given** a student has an active checked-in reservation, **When** the student checks out, **Then** the system records the check-out time and releases the seat for subsequent eligible bookings.

US-11: Monitor Occupancy

- **Given** seat reservations and check-in records exist, **When** a librarian opens the occupancy dashboard, **Then** the system displays the latest recorded seat and reservation statuses.
- **Given** a seat's status changes, **When** the change is saved, **Then** the dashboard reflects the updated status.

US-12: Detect Unused Reservations

- **Given** a reservation has passed its configured check-in grace period without a valid check-in, **When** the no-show detection process runs, **Then** the system marks the reservation as a no-show and releases the seat according to library policy.
- **Given** a student has checked in within the allowed period, **When** the detection process runs, **Then** the system does not incorrectly release that reservation.

US-13: Enforce Reservation Time Limits

- **Given** an administrator has configured maximum booking durations and grace periods, **When** a student attempts a reservation, **Then** the system validates the requested time against those limits.
- **Given** a reservation reaches its end time, **When** the configured expiry process runs, **Then** the system updates the reservation according to the defined policy.

US-14: Reservation Notifications

- **Given** a reservation is successfully created, **When** the booking is confirmed, **Then** the system generates a confirmation notification.
- **Given** a reservation reminder is due, **When** the notification process runs, **Then** the system attempts to send the reminder through the configured notification service.

US-15: Seat Utilization Reports

- **Given** reservation and occupancy records are available, **When** an administrator selects a reporting period, **Then** the system generates a report containing total seats, recorded occupied time, available time, and utilization rate.
- **Given** no records exist for the selected period, **When** the report is requested, **Then** the system displays an appropriate empty-state message.

4. Prioritize the Product Backlog

The product backlog is an ordered list of features and improvements required for the system. The backlog is prioritized using business value, student needs, dependencies, and implementation importance.

Priority levels are defined as:

- **High:** Essential for the core reservation workflow.
- **Medium:** Improves management and usability after core functionality is available.
- **Low:** Can be developed after the essential features.

Priority	Story ID	User Story	Reason
1 – High	US-01	Student login	Provides secure access to the system.
2 – High	US-03	View available seats	Essential for finding study spaces.
3 – High	US-04	View seat details	Helps students select suitable seats and time slots.
4 – High	US-06	Reserve a seat	Implements the primary business function.
5 – High	US-07	Prevent double booking	Ensures reservation correctness.
6 – High	US-08	Cancel a reservation	Allows seats to be reused when plans change.
7 – High	US-09	Check in	Records student arrival.
8 – High	US-10	Check out	Releases seats after use.
9 – Medium	US-05	Manage seat information	Supports administrative maintenance.
10 – Medium	US-11	Monitor occupancy	Improves operational visibility.
11 – Medium	US-12	Detect unused reservations	Reduces seat wastage.
12 – Medium	US-13	Enforce time limits	Supports consistent reservation policies.
13 – Medium	US-14	Reservation notifications	Improves communication and convenience.
14 – Medium	US-02	Role-based access	Strengthens administrative access control.
15 – Low	US-15	Seat utilization reports	Supports longer-term planning and analysis.

Prioritization justification: Login, seat availability, and reservation are placed first because they establish the main user journey. Double-booking prevention is essential for data integrity. Check-in and check-out follow to support seat usage tracking. No-show detection depends on reservation and check-in records, while utilization reports require sufficient data to produce meaningful results.

Security controls remain necessary throughout development, even when advanced administrative permissions are scheduled for a later sprint.

5. Plan the Sprint Backlog

5.1 Sprint Details

Sprint Number: Sprint 1

Duration: 2 weeks

Sprint Theme: Basic Student Access and Seat Reservation

Sprint Objective: Develop a working initial version that allows students to log in, view available seats, inspect time slots, and reserve seats without conflicting bookings.

5.2 Selected User Stories

Story ID	User Story	Priority	Story Points
US-01	Student login	High	3
US-03	View available seats	High	3
US-04	View seat details	High	2
US-06	Reserve a seat	High	5
US-07	Prevent double booking	High	5
Total	**5 user stories**		**18**

The team commits to these stories only if the estimated 18 points fit its capacity. The actual sprint commitment should be confirmed after considering team size, availability, and previous velocity.

5.3 Actionable Development Tasks

US-01: Student Login — 3 Story Points

1. Define the student account fields in the database.
2. Design the login page.
3. Implement credential validation and secure password verification.
4. Add session management and logout functionality.
5. Display appropriate success and error messages.
6. Test valid and invalid login attempts.

Expected result: Registered students can securely access the application.

US-03: View Available Seats — 3 Story Points

1. Create the seat database table.
2. Define seat statuses and study-area information.
3. Implement the backend endpoint or function to retrieve seats.
4. Design the seat availability page.
5. Display seat identifiers and availability status.
6. Test the display of available, reserved, and unavailable seats.

Expected result: Students can view current seat availability.

US-04: View Seat Details — 2 Story Points

1. Design the seat details interface.
2. Display seat location and identifier.
3. Retrieve available time slots from the backend.
4. Validate that the displayed slots are suitable for booking.
5. Test the seat details and time-slot display.

Expected result: Students can inspect a seat and select a valid time slot.

US-06: Reserve a Seat — 5 Story Points

1. Design the reservation form.
2. Create the reservation database table.
3. Implement seat and time-slot validation.
4. Implement reservation creation.
5. Update the displayed availability after successful booking.
6. Display a reservation confirmation.
7. Test successful, invalid, and unavailable-seat booking attempts.

Expected result: Students can reserve available seats for valid time slots.

US-07: Prevent Double Booking — 5 Story Points

1. Define rules for overlapping reservation periods.
2. Implement server-side availability checks.
3. Use database transactions to process reservation requests safely.
4. Add suitable database constraints where applicable.
5. Handle simultaneous booking attempts.
6. Display a conflict message when a seat is already booked.
7. Test overlapping and simultaneous reservation requests.

Expected result: The system prevents conflicting confirmed reservations for the same seat.

5.4 Sprint Task Summary

Task Category	Main Activities
Analysis	Confirm requirements and reservation rules.
Database	Create student, seat, and reservation tables.
Interface	Build login, availability, details, and booking pages.
Backend	Implement authentication, seat retrieval, and reservation operations.
Validation	Add time-slot validation and conflict prevention.
Testing	Perform unit, integration, and acceptance tests.
Review	Demonstrate the completed features and collect feedback.

5.5 Definition of Done

A user story is considered complete when:

1. All acceptance criteria are satisfied.
2. The implementation follows the agreed coding standards.
3. Unit and integration tests pass.
4. Relevant validation and error handling are implemented.
5. No known critical defect remains in the completed functionality.
6. Changes are reviewed and integrated into the shared codebase.
7. The feature is demonstrated successfully during the sprint review.
8. Necessary technical documentation is updated.

6. Estimate Effort Using Story Points

6.1 Estimation Technique

The team uses Planning Poker with the modified Fibonacci sequence: 1, 2, 3, 5, 8, and 13.

Story points represent relative effort, complexity, uncertainty, and testing needs. They do not represent a fixed number of hours or days.

During Planning Poker, each team member estimates a story independently. The team discusses differences in estimates and agrees on a final value.

6.2 Story Point Estimates

Story ID	User Story	Story Points	Justification
US-01	Student login	3	Requires interface development, credential verification, and session handling.
US-03	View available seats	3	Requires seat retrieval, status display, and availability testing.
US-04	View seat details	2	Relatively straightforward display and time-slot retrieval functionality.
US-06	Reserve a seat	5	Requires reservation validation, database operations, and status updates.
US-07	Prevent double booking	5	Requires transaction handling, conflict detection, and concurrency testing.
Total		**18**	

6.3 Explanation of the Fibonacci Scale

- **1 point:** Very simple task with minimal complexity.
- **2 points:** Small task with limited validation.
- **3 points:** Moderate task involving several related operations.
- **5 points:** More complex task requiring multiple components and thorough testing.

- **8 points:** Large or uncertain task that may need further breakdown.
- **13 points:** Very large task that should normally be divided into smaller stories.

The estimates are preliminary and should be validated by the development team. If the team is new to the project, its capacity should be assessed carefully before committing to the entire backlog.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

To deliver a working foundation for the Smart Library Seat Reservation and Study Space Management System in which registered students can log in, view available seats and time slots, and reserve a seat without conflicting bookings.

7.2 Expected Deliverables

At the end of Sprint 1, the team expects to deliver:

1. A functional student login and logout interface.
2. A database structure for student accounts, seats, and reservations.
3. A seat availability page displaying current seat statuses.
4. A seat details page showing identifiers and bookable time slots.
5. A working reservation form.
6. Backend reservation validation and confirmation.
7. Protection against overlapping and simultaneous conflicting bookings.
8. Unit and integration test results for the completed features.
9. Basic technical documentation and a sprint review demonstration.

7.3 Sprint Acceptance Conditions

The sprint is considered successful when:

- A registered student can log in and log out.
- Seat availability is displayed correctly from the database.
- Students can inspect available time slots.
- A valid reservation is saved and confirmed.
- An unavailable seat cannot be booked for a conflicting time slot.
- Simultaneous booking attempts cannot create conflicting confirmed reservations.
- The completed stories satisfy their acceptance criteria and Definition of Done.

7.4 Sprint Review

At the end of the sprint, the development team demonstrates the working application to the product owner, librarians, and other relevant stakeholders. Feedback is collected on usability, booking rules, and missing functionality.

Any new requirements or improvements are added to the product backlog and prioritized for upcoming sprints.

7.5 Sprint Retrospective

The development team discusses what went well, what problems occurred, and what improvements should be made in the next sprint.

Possible discussion points include:

- Whether story point estimates were realistic.
- Whether the database design supported the required features.
- Whether testing detected booking conflicts early enough.
- Whether team communication and task allocation were effective.
- What changes could improve development quality and speed.

Conclusion

The Agile Sprint Planning and User Story Workshop provides a structured approach to developing the Smart Library Seat Reservation and Study Space Management System. The project vision identifies the need for efficient study-space management, while epics and user stories divide the requirements into manageable features.

Acceptance criteria establish measurable conditions for completion, and product backlog prioritization ensures that essential functions are developed first. Sprint 1 focuses on student authentication, seat availability, seat details, reservation, and double-booking prevention.

The use of Fibonacci story points, a clear sprint goal, actionable tasks, and a Definition of Done supports effective teamwork and quality development. Future sprints can introduce check-in and check-out, occupancy monitoring, automatic no-show detection, notifications, and utilization reports. Through iterative development and continuous stakeholder feedback, the system can improve student convenience and library resource utilization.